



Top 100 DevOps Engineer Interview Questions & Answers

1. DevOps Fundamentals

1. What is DevOps?

DevOps is a software development methodology that integrates development (Dev) and operations (Ops) to improve collaboration, automation, and efficiency.

2. What are the key principles of DevOps?

- Collaboration
- Automation
- Continuous Integration & Deployment
- Monitoring & Feedback
- Security & Compliance

3. How does DevOps differ from Agile?

Agile focuses on software development processes, while DevOps extends it to operations, ensuring faster and reliable delivery.

4. What are the key benefits of DevOps?

- Faster software releases
- Improved collaboration
- Higher efficiency and scalability
- Better system reliability

5. What are the key DevOps tools?

- CI/CD: Jenkins, GitLab CI, GitHub Actions
- Containerization: Docker, Podman
- Orchestration: Kubernetes, OpenShift
- Monitoring: Prometheus, Grafana
- Configuration Management: Ansible, Puppet, Chef
- Version Control: Git

2. Version Control & Git

6. What is Git?

Git is a distributed version control system for tracking source code changes.

7. Difference between Git and GitHub/GitLab?

Git is a VCS, whereas GitHub/GitLab are web-based platforms providing repositories with collaboration features.

8. What is the difference between git pull and git fetch?

- git fetch: Downloads changes but doesn't merge.
- git pull: Fetches and merges changes into the working branch.

9. Explain Git branching strategies.

- **Feature Branching:** Develop features in isolated branches.
- **Gitflow:** Uses main, develop, feature, release, and hotfix branches.
- **Trunk-based development:** Continuous integration into main.

10. How do you resolve a merge conflict in Git?

- Identify conflicts using git status.
- Manually edit the conflicting files.
- Add (git add) and commit (git commit -m "Resolved conflict") the resolved files.

3. CI/CD (Jenkins, GitLab CI/CD, GitHub Actions)

11. What is Continuous Integration (CI)?

CI automates code integration from multiple developers into a shared repository.

12. What is Continuous Deployment (CD)?

CD automates software delivery from testing to production.

13. Explain Jenkins Pipeline.

- A **Declarative Pipeline** defines the entire CI/CD process in Jenkinsfile.

- A **Scripted Pipeline** provides greater flexibility but requires Groovy scripting.

14. How do you secure Jenkins?

- Use Role-Based Access Control (RBAC).
- Encrypt secrets using Jenkins credentials.
- Use HTTPS and limit plugin vulnerabilities.

15. What are GitHub Actions and their advantages?

- A CI/CD automation tool integrated with GitHub.
- Advantages: Easy setup, built-in marketplace, YAML-based workflows.

4. Configuration Management (Ansible, Puppet, Chef)

16. What is Configuration Management in DevOps?

Managing system configurations to ensure consistency and scalability.

17. How does Ansible differ from Puppet and Chef?

- **Ansible:** Agentless, YAML-based, push model.
- **Puppet:** Agent-based, uses Puppet DSL, pull model.
- **Chef:** Uses Ruby DSL, agent-based.

18. What is an Ansible Playbook?

A YAML file defining automation tasks.

19. Explain Infrastructure as Code (IaC).

Automating infrastructure provisioning using code (e.g., Terraform, Ansible).

20. What is an Ansible role?

A structured way to organize Ansible Playbooks.

5. Containers & Orchestration (Docker, Kubernetes)

21. What is Docker?

Docker is a containerization platform that packages applications with dependencies.

22. What is a Dockerfile?

A script containing instructions to build a Docker image.

23. What is Kubernetes?

An orchestration platform for managing containerized applications.

24. What are Kubernetes Pods?

The smallest deployable unit in Kubernetes, containing one or more containers.

25. What is Helm in Kubernetes?

A package manager for Kubernetes that simplifies application deployment.

6. Cloud & Infrastructure as Code (AWS, Azure, GCP, Terraform)

26. What is Terraform?

An open-source IaC tool for managing cloud infrastructure declaratively.

27. What is the difference between Terraform and CloudFormation?

- Terraform: Multi-cloud, state management, declarative.
- CloudFormation: AWS-specific, integrated with AWS services.

28. Explain the Terraform state file.

Stores infrastructure state to track changes.

29. What is an AWS IAM role?

A set of permissions for AWS services to access resources securely.

30. What is auto-scaling in AWS?

Automatically adjusting the number of instances based on demand.

7. Monitoring & Logging (Prometheus, Grafana, ELK Stack)

31. What is Prometheus?

An open-source monitoring system for collecting and querying time-series data.

32. What is Grafana used for?

A visualization tool for monitoring metrics.

33. Explain the ELK Stack.

- **Elasticsearch:** Search engine
- **Logstash:** Log ingestion
- **Kibana:** Visualization

34. What is observability?

The ability to measure system health via logs, metrics, and traces.

35. What are service-level objectives (SLOs)?

Targets for system performance and availability.

8. Security in DevOps (DevSecOps)

36. What is DevSecOps?

Integrating security into DevOps workflows.

37. What is OWASP?

Open Web Application Security Project – provides security guidelines.

38. How do you secure a containerized environment?

- Use minimal base images.

- Implement RBAC in Kubernetes.
- Scan images for vulnerabilities.

39. What is Shift-Left Security?

Incorporating security early in the software development lifecycle.

40. What are secrets management tools?

HashiCorp Vault, AWS Secrets Manager, Kubernetes Secrets.

9. SRE & Reliability Engineering

41. What is Site Reliability Engineering (SRE)?

A discipline that applies software engineering to infrastructure operations.

42. What are SLAs, SLOs, and SLIs?

- **SLA:** Service Level Agreement
- **SLO:** Service Level Objective
- **SLI:** Service Level Indicator

43. What is Chaos Engineering?

The practice of testing system resilience through controlled failures.

44. What is an Error Budget?

The acceptable downtime limit before affecting SLOs.

45. What are blameless postmortems?

Incident reviews focused on learning rather than blaming.

10. Advanced CI/CD Concepts

46. How do you implement Blue-Green Deployment?

- Maintain two environments (Blue = live, Green = new).
- Switch traffic after testing Green.

47. What is Canary Deployment?

- Gradual release to a small subset of users before full rollout.

48. What is a Rolling Update?

- Gradual replacement of old instances with new ones without downtime.

49. How do you handle secrets in CI/CD pipelines?

- Use environment variables, HashiCorp Vault, or AWS Secrets Manager.

50. How do you prevent deployment failures?

- Implement automated testing, rollback strategies, and feature flags.

11. GitOps & Infrastructure Automation

51. What is GitOps?

- A DevOps model where infrastructure changes are managed via Git repositories.

52. How does GitOps differ from traditional IaC?

- GitOps enforces version-controlled infrastructure and automatic reconciliation.

53. What are the best practices for Terraform state management?

- Store state in remote backends (S3, Azure Blob).
- Use state locking to prevent conflicts.

54. What is a Terraform module?

- A reusable, parameterized collection of Terraform configurations.

55. What is drift detection in Terraform?

- Detecting changes in infrastructure that are not in the Terraform state.

12. Kubernetes Advanced Topics

56. What is a Kubernetes DaemonSet?

- Ensures a pod runs on every node in a cluster.

57. What is a Kubernetes StatefulSet?

- Used for stateful applications, providing stable network IDs and persistent storage.

58. How does Kubernetes Horizontal Pod Autoscaler (HPA) work?

- Adjusts the number of pods based on CPU/memory metrics.

59. What is a Kubernetes Ingress?

- A resource managing external access to services via HTTP/HTTPS.

60. What is Kubernetes RBAC?

- Role-Based Access Control for managing permissions in a cluster.

13. Monitoring & Logging Advanced Concepts

61. What is PromQL in Prometheus?

- A query language for fetching Prometheus metrics.

62. How do you monitor Kubernetes clusters?

- Use Prometheus, Grafana, and Kubernetes Metrics Server.

63. How do you centralize logs in a distributed system?

- Use the ELK Stack or Fluentd for log aggregation.

64. What is the difference between tracing and logging?

- Logging captures discrete events; tracing follows a request's lifecycle.

65. How does OpenTelemetry help with observability?

- Provides unified telemetry (logs, metrics, traces) across services.

14. Security Best Practices in DevOps

66. How do you implement DevSecOps in a pipeline?

- Integrate security scanning tools (SAST, DAST) into CI/CD.

67. What is a SAST tool?

- Static Application Security Testing (e.g., SonarQube, Snyk).

68. What is a DAST tool?

- Dynamic Application Security Testing (e.g., OWASP ZAP, Burp Suite).

69. How do you implement least privilege access in DevOps?

- Use IAM roles, RBAC, and enforce MFA.

70. What is container image scanning?

- Scanning Docker images for vulnerabilities using tools like Trivy or Clair.

15. Cloud-Native & Serverless

71. What is a serverless architecture?

- Running applications without managing infrastructure (e.g., AWS Lambda).

72. What are the benefits of serverless computing?

- Auto-scaling, cost efficiency, and reduced operational overhead.

73. How does Kubernetes compare to AWS Lambda?

- Kubernetes runs containerized apps, whereas Lambda is event-driven and serverless.

74. What is a CloudFormation Stack?

- A collection of AWS resources managed as a single unit.

75. What is an API Gateway in cloud environments?

- A managed service for routing, securing, and monitoring API requests.

16. Advanced Networking & Security

76. What is a Service Mesh?

- A dedicated infrastructure layer for managing service-to-service communication (e.g., Istio, Linkerd).

77. How do you secure microservices communication?

- Use TLS encryption, API gateways, and mutual TLS authentication.

78. What is a Zero Trust security model?

- A model where no one is trusted by default, requiring strict identity verification.

79. What is a WAF (Web Application Firewall)?

- Protects applications from web-based threats like SQL injection and XSS.

80. How do you protect against DDoS attacks?

- Use CDNs, rate limiting, and AWS Shield/Cloudflare protections.

17. DevOps Culture & Processes

81. How do you implement DevOps in a large enterprise?

- Start with CI/CD adoption, IaC, monitoring, and DevSecOps practices.

82. What are key DevOps KPIs?

- Deployment frequency, mean time to recover (MTTR), change failure rate.

83. How do you handle failures in a DevOps environment?

- Implement rollback strategies, blameless postmortems, and chaos engineering.

84. What is a postmortem in DevOps?

- A retrospective analysis of an incident to prevent recurrence.

85. What are feature flags?

- A mechanism for toggling features on/off in production.

18. Performance Optimization

86. What is a CDN, and why is it used?

- A Content Delivery Network caches content to reduce latency.

87. How do you optimize CI/CD pipeline performance?

- Use parallel builds, caching, and selective testing.

88. How do you optimize database performance in DevOps?

- Indexing, caching, and database partitioning.

89. What is a sidecar pattern in microservices?

- Deploying an auxiliary container alongside the main service for logging, monitoring, or security.

90. How do you reduce cloud costs in a DevOps environment?

- Use auto-scaling, spot instances, and cost monitoring tools.

19. Incident Management & Disaster Recovery

91. What is a runbook in DevOps?

- A predefined set of procedures for handling incidents.

92. What is a playbook in incident response?

- A detailed action plan for mitigating security or system issues.

93. How do you handle rollback in Kubernetes?

- Use kubectl rollout undo to revert to the previous deployment.

94. What is RTO and RPO in disaster recovery?

- RTO: Recovery Time Objective (time to restore services).
- RPO: Recovery Point Objective (maximum acceptable data loss).

95. How do you test disaster recovery in DevOps?

- Conduct failover tests and simulate outages.

20. Miscellaneous & Future Trends

96. What is FinOps in cloud computing?

- Financial operations to optimize cloud spending.

97. What is Chaos Engineering?

- Deliberate testing of system failures to improve resilience.

98. What is Policy-as-Code?

- Defining security and compliance policies in code (e.g., OPA, AWS SCPs).

99. What is AIOps?

- AI-driven operations that automate incident detection and resolution.

100. What are emerging trends in DevOps for 2025?

- AI-driven automation, GitOps adoption, enhanced Kubernetes security, and observability improvements.